

AI for Agriculture: Towards Digital Twins for Methane Emissions Forecasting and Scenario Analysis

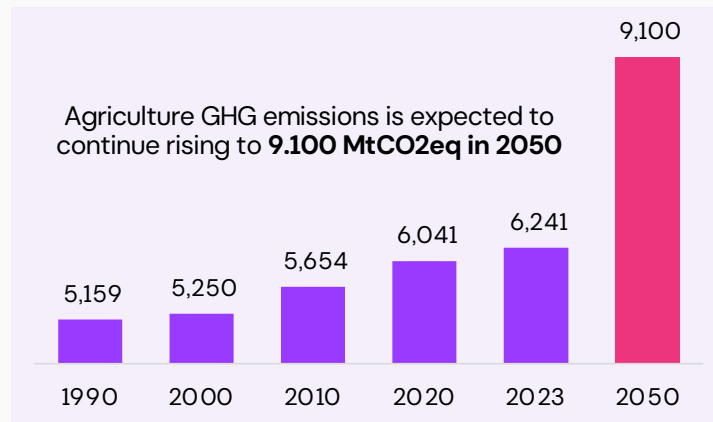
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25/03/2026

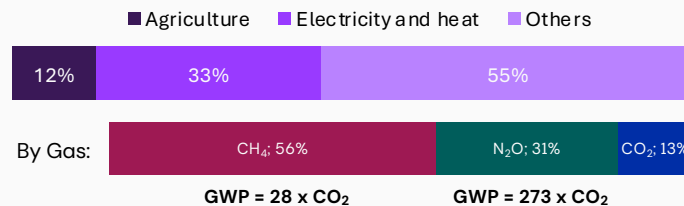
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02 | Background



Agriculture represents 12% of emissions worldwide. Most coming from CH₄ and N₂O where Agriculture is #1.



Agricultural emissions is an active research topic

Purcell & Neubauer (2023) paper on state of agricultural digital twins (DT) found that what-if scenario simulation remains critically underexplored. In which, DT's related to emissions are even rarer.

► Key Existing Research:

- DT framework at farm scale proposed (*Fakeye et al., 2024*), emissions as a target but still a framework only.
- Closest related ML work: GreenFeed-based cattle CH₄ prediction (*Partridge & Alhnaity, 2024*), EC data mostly unused for prediction.
- Process based models (*Wu et al., 2022*) explored to simulate CH₄. But are static, offline, and require fixed inputs.

► Unresolved Problems & Challenges:

- NWFP flux monitoring (Eddy Covariance) produces 7+ years of high frequency data but is largely unused in forecasting. (*Zhu et al, 2023*)
- EC outputs are driven by multiple drivers (enteric fermentation, soil moisture, fertilizer, and others) simultaneously and are complex to model.
- **No system integrates continuous EC CH₄ flux with temporal forecasting for farm-scale emission prediction and scenario analysis.**

03 | Aims & Objectives

AIM

The development and evaluation of multiple ML approaches for CH₄ forecasting at North Wyke Farm Platform and demonstrate its integration with a digital shadow architecture with scenario analysis and uncertainty quantification.

Objectives

1. Conduct **further systematic literature review** on time series forecasting, quantile machine learning, EC sensors, etc.
2. Acquire, **preprocess NWFP data** and perform rigorous data handling + imputation
3. **Pipelining** (Feature Engineering, Model Selection and Assessment) and **benchmarking for multiple models**.
4. **Interpretability and Uncertainty Quantification** – driven by permutation importance and quantile machine learning (*Oulaid & Harris et al., 2025*)
5. **Scenario Generation** (dictated by feature engineering and farm management practices) and **model testing** for In sample and out of sample scenarios
6. **Creation of scenario-analysis digital shadow** informed by the predictive model on Streamlit / WebApp

Hypothesis

1. EC will capture a more grounded and accurate emissions prediction compared to IPCC Tier 1 Static Emission Factors
2. ML-based digital shadow will produce multi-step emission forecasts and statistically distinguishable, interpretable predictions under contrasting management (known) & climate (unknown) scenarios.

04 | Approach

Data & Sources

- ▶ **Dataset: Rothamsted NWFP (Open-Source, Three Catchments)**
 - ▶ Main Dataset: EC GHG flux for CH₄ (30 minute resolution, 7+ years data, gaps in dataset), Farm management (manual)
 - ▶ Complementing Datasets: Meteorological, Soil moisture, temperature, rainfall data (15 minute resolution)
- ▶ **Expected Data Pre-processing**
 - ▶ Data imputation, feature engineering, data splitting

Expected Outcomes and Deliverables

- ▶ Best-performing model identified based on performance and runtime considerations
- ▶ Statistically distinguishable and meaningful scenario predictions
- ▶ Interpretable model driven by SHAP and other methods
- ▶ Functional digital shadow webapp

Methodology & Approach:

- ▶ Model comparison for three farm types:
 - ▶ ARIMA, Bagging / Boosting, LSTM, Transformer, and more
- ▶ Interpretability and Uncertainty Quantification:
 - ▶ Global & Local Interpretability and Prediction Intervals
- ▶ Scenario Analysis:
 - ▶ Synthetic management scenarios and digital shadow interface on Streamlit / webapp

How will success be measured?

- ▶ **Based on temporal cross-validation evaluation**
 - ▶ Example: train (2018 – 2021), test (2022 – 2023)
 - ▶ Metrics:
 - ▶ Regression: RMSE, MAE
 - ▶ Uncertainty: Coverage Probability and Interval Score
 - ▶ Interpretability: Model adherence and plausibility based on the domain

05 | AI Regulatory Considerations

Cross-Sectoral Principles

Safety

- Wrong predictions equals wrong management practices.
- Mitigated by Uncertainty.

Fairness

- Model biased for grassland and livestock farming in the UK & Western Europe.

Transparency

- Supported by uncertainty, interpretability, and DT component.

Accountability

- Farmers and researchers as decision makers.
- DT as support tool only

Contestability

- Interpretability component supports contestability.
- Further supported by DT and Scenario Analysis component.

AI System Classification

- Based on dataset and application: **Narrow AI**
 - Model will NOT autonomously perform sequences of actions (~~agent~~) and are NOT adaptable across domains (~~foundational~~)
 - Proposed project will be done on a domain specific dataset for a domain specific task (emissions prediction).

Societal Harm & Misuse Risks

Potential Risks:

- Over-reliance on prediction
- Unintentional Feedback Loop
- Intrinsic bias from geography
- Misuse for greenwashing

Mitigation Strategy:

- Probabilistic prediction w/ intervals
- System must only inform, never decide
- Explicit scope should be highlighted
- Uncertainty component